

sp(eye)

Software Engineering Group Project

Group 02 – Alexander Chow, Caleb Kan, Jeffrey Cheung, Kee Meng Tan, Royce Li Wei Chan, Ryan Ghoussainy



1 in 5



Britons aged 16–65 can only
read **at or below** the level
expected of a 10-year-old.



50%

of UK adults **don't** regularly read

1 in 4

young people (16-24) say
they've **never** been readers



238
words per minute (WPM)

The current average adult reading speed.
This falls roughly **20% slower** than the
widely assumed **300 WPM benchmark**.

User Needs

Primary

Read faster while
maintaining comprehension

Target Users

Anyone who wishes to
learn how to read faster

Secondary

- Read at a fixed, configurable WPM
- Read with adaptive WPM with eye-tracking
- Read from a library of public texts
- Upload and read their own texts
- Gauge comprehension with quizzes
- Track past activity to see progress over time

Standard Mode

Adaptive Mode

Library

Quizzes

User Stats

Admin

Standard Mode

10–15%

of skilled readers make regressions, moving backward in the text to previous word



So Much to Read, So Little Time: How Do We Read, and Can Speed Reading Help?

Keith Rayner^{1†}, Elizabeth R. Schotter¹, Michael E. J. Masson², Mary C. Potter³, and Rebecca Treiman⁴

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Psychological Science in the Public Interest
2016, Vol. 17(1) 4–34
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sagepub.com/journalsPermissions.nav
DOI: 10.1177/1529100615623267
pspi.sagepub.com
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<https://journals.sagepub.com/doi/10.1177/1529100615623267>

Adaptive Mode

Gaze-Driven Reading

Webcam + WebGazer

→ No fixed speed setting, you read at your natural pace

Calibration:

9-point click calibration + accuracy test (>50%)

→ Stored locally (no need to redo)

Gaze Processing Pipeline

Raw gaze (60 FPS)

→ Smoothing, outlier rejection, confidence scoring.

Normalisation layers drives the reading experience

Visual feedback

Border/gradient colours

→ change based on tracking state and gaze position
(approaching end, in end zone, sweep detected)

Text is shown one line at a time

Return Sweep detection: Detects when your eyes sweep from right-to-left at the end of the line (velocity transition detection), then auto-advances to the next chunk of text.

Radio 1 (top 5) and found on the OCC website as a Top 100

Chunk 7 of 24

● Tracking

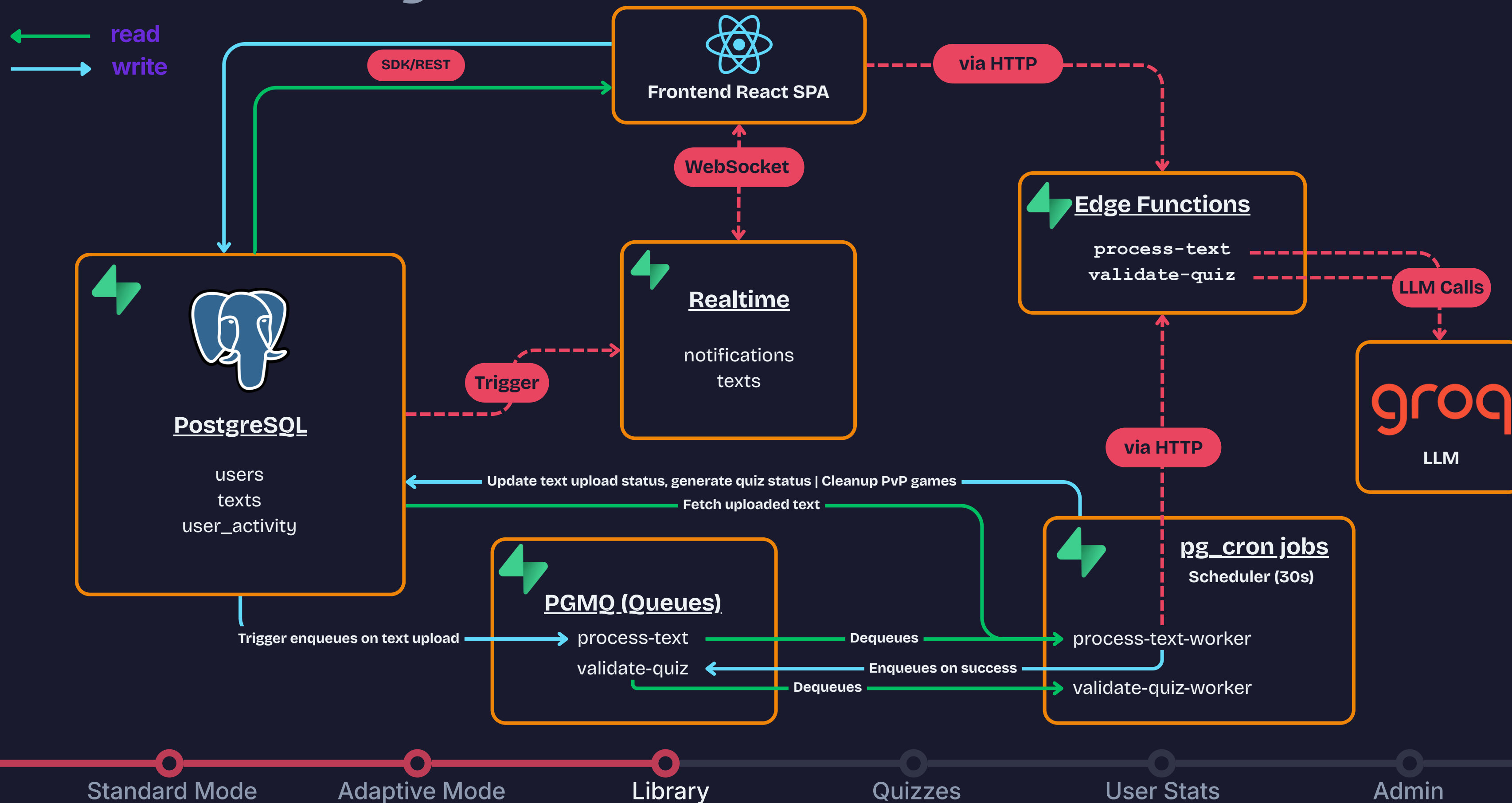
Radio 1 (top 5) and found on the OCC website as a Top 100

Chunk 7 of 24

● End zone

continuing until all sales have been tracked in data only

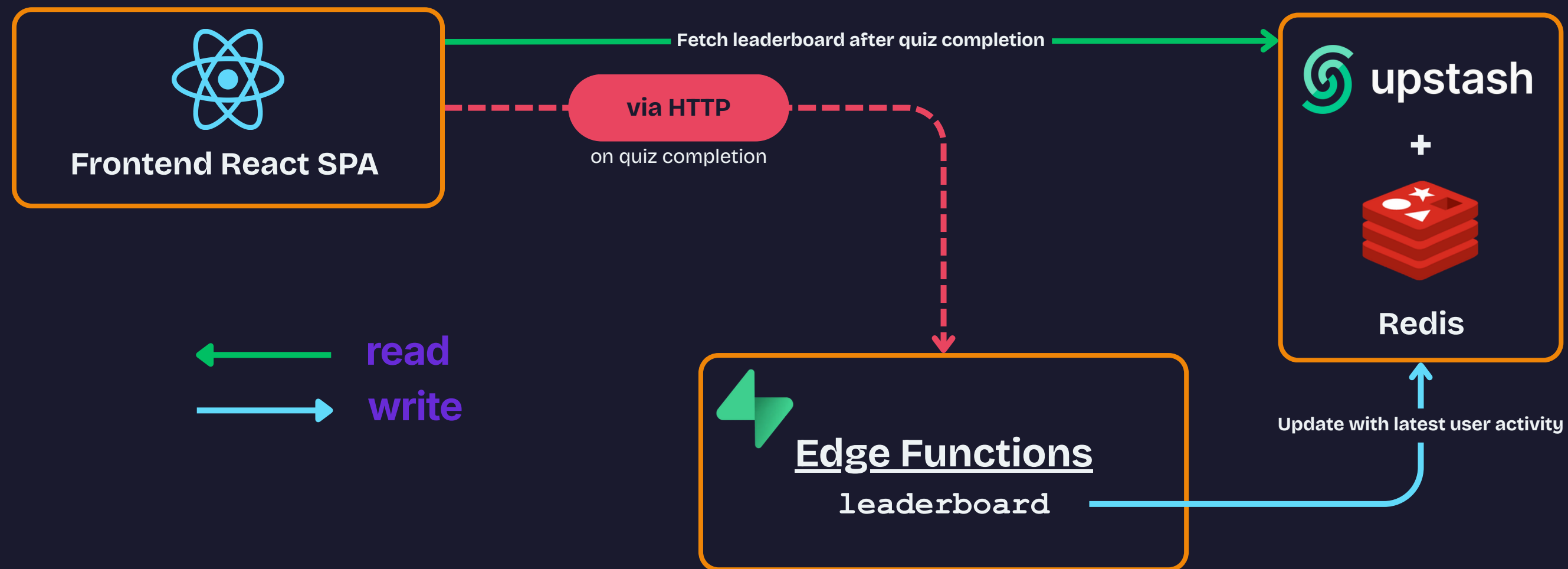
Library + Summarised Mode



Quizzes

Why Redis?

- User activity table origin
- New row inserted every time a user changes speed/changes mode
- Lots of rows
- Need millisecond reads (users unlikely to stay on modal for long)



Quizzes

$$\text{Overall Score} = \left[\text{WPM}^{0.7} \times \left(\frac{A - 0.25}{0.75} \right)^{1.5} \times 10 \right]$$

where:

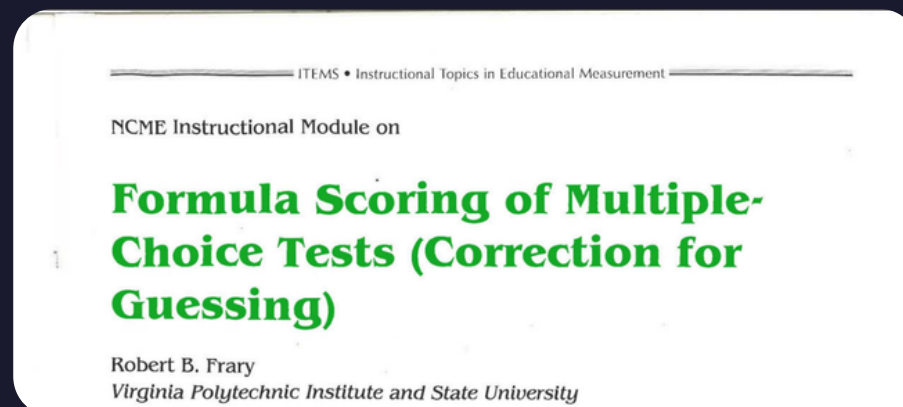
WPM = user's average wpm

A = quiz score

Chance-correction floor at 25%
neutralises random guessing on
questions (given 4 options)

The super-linear accuracy exponent (1.5) penalises
poor comprehension more than raw speed

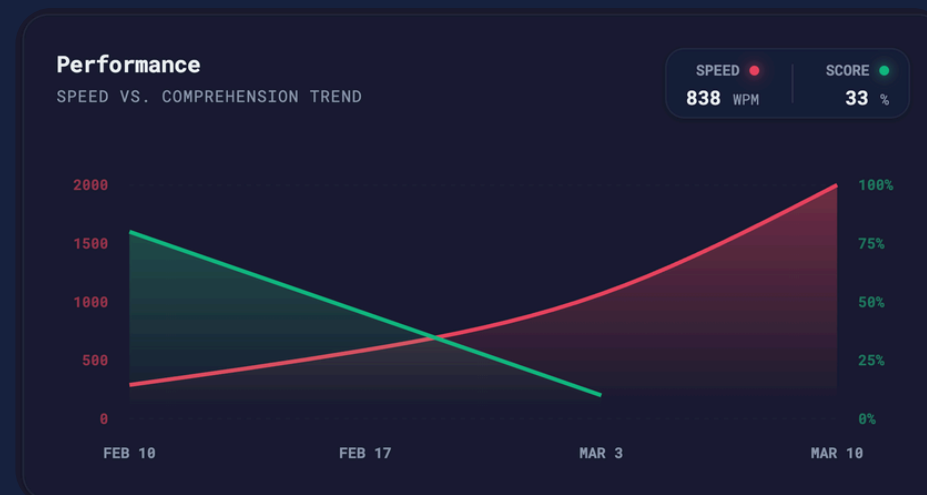
Sub-linear speed exponent (0.7) reflects
diminishing returns beyond a reader's
comfortable pace



User Stats + Activity

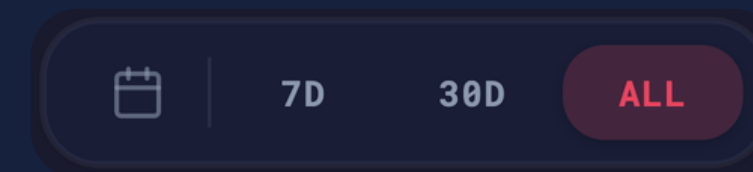
Key Metrics

- Average Speed (WPM)
- Average Score (%)
- Texts Read
- Current + Best Streak



+ Trends

ActivityHeader.tsx



ActivityHeader → `invokeRPC()`

adminService.ts (interface)

```
WpmStatsParams  
{ timeframe: "last_7d", passing_threshold: 25 }
```

API Gateway (Supabase)

{POST}

`get_admin_wpm_distribution`



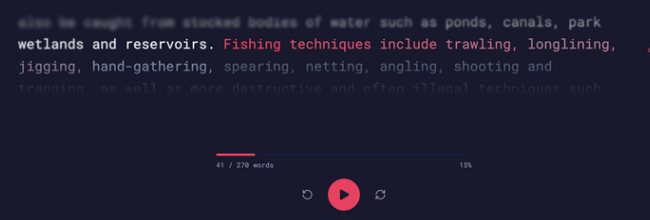
Activity

Segments



Last Saved Index

Fishing



Standard Mode

Adaptive Mode

Library

Quizzes

User Stats

Admin

Admin Panel

Key Metrics

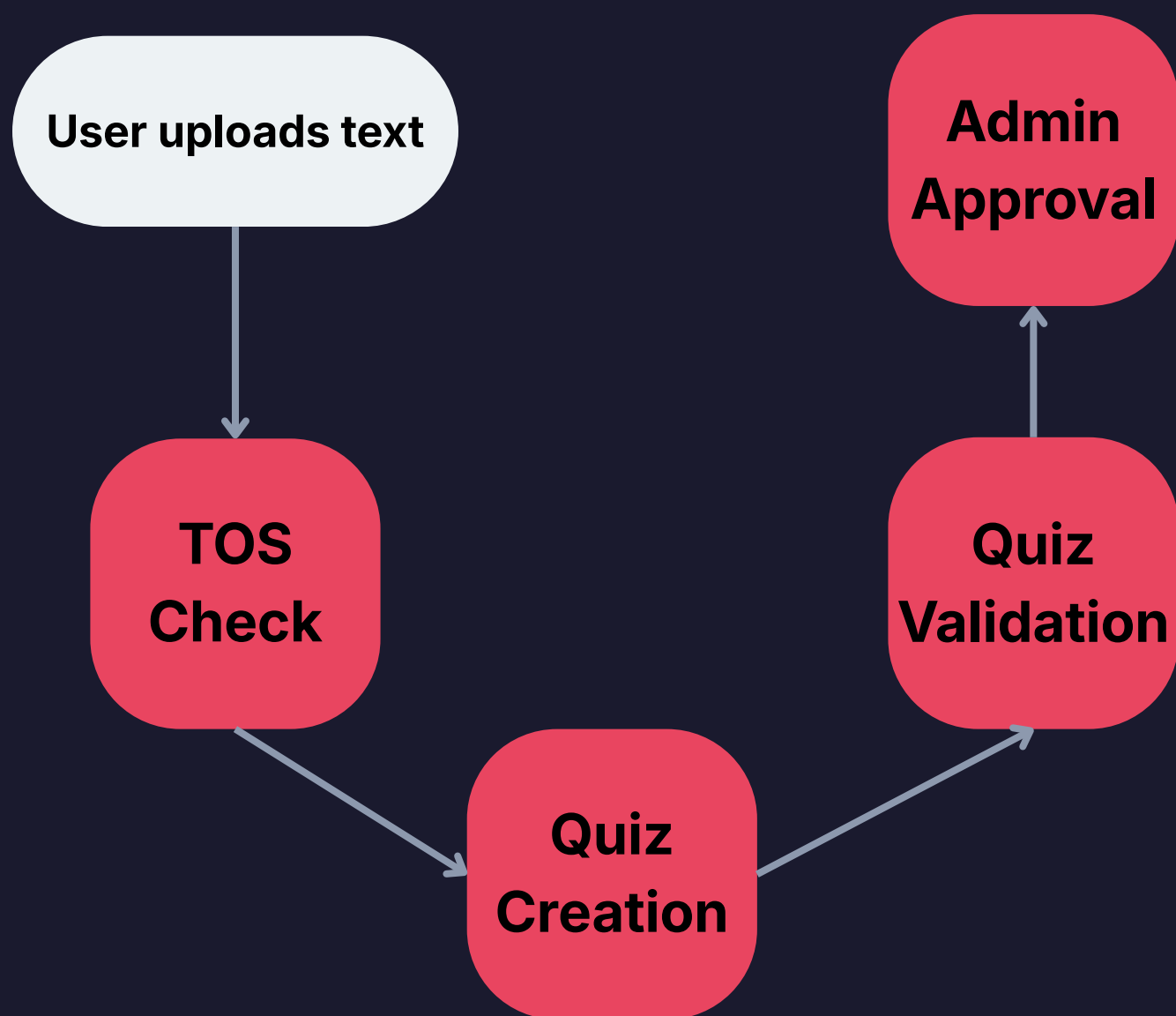
Global Stats

- Average Speed (WPM)
- Average Score (%)
- Total Texts
- Rejection Rate (%)



Admin Panel

Text Approval + Rejection



Notifications

Sending

Insert into 'notifications' table



Receiving

Subscribe to realtime updates to 'notifications' table targetted at you

Admin Promotion

Username Search

Standard Mode

Adaptive Mode

Library

Quizzes

User Stats

Admin

Valuable Extensions

Users loved the platform but wanted more reasons to come back

→ **Introduced PvP Mode:** 1v1 competitive reading with Elo ranking

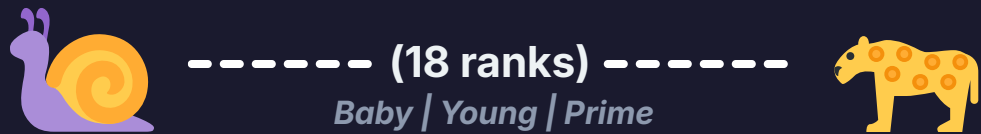
Users wanted to read on the go, not just on laptops

→ Introduced mobile support with RSVP (Rapid Serial Visual Presentation) mode
(optimised for small screens)

→ Enabled offline operations via PWA (Progressive Web Application)

PvP

Elo Rankings



Real-Time Sync

Supabase Realtime Channels

- Broadcast progress, milestones, and heartbeats.
- **Reliable**: Polling fallback every 2-10s

Fair Timing

Clock offset: server RPC + Round-Trip Time measurement

- Allows both players to see the same countdown.

Resilience

AFK auto-forfeit (30 seconds)

- *keepalive* fetch for tab-close forfeit
- *sessionStorage* crash recovery for quiz submissions
- Exponential backoff reconnection to Supabase realtime channels

Scoring

- Overall score formula from quizzes
- Elo adjusted accordingly

Elo-based matchmaking pairs players of similar skill


woka
884
8W - 18L

VS


woka1106
1086
15W - 4L

Both players read the same text and take the same quiz

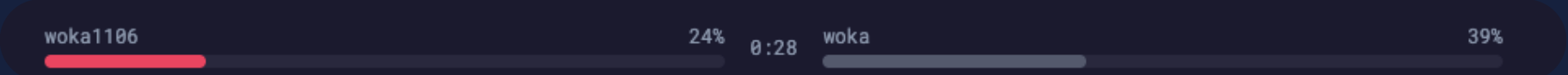
...differences of words found throughout the German speaking world. The dictionary's historical linguistics approach, illuminated by examples from primary source documents, makes it to German what the Oxford English Dictionary is to English. The first completed DWB lists over 330,000 headwords in 67,000 print columns spanning 32 volumes. The Deutsches Wörterbuch was begun by the Brothers Grimm in 1838 and the initial volumes were published in 1854. Unfinished at the time of their deaths, the dictionary was finally completed by a succession of

Comprehension Quiz 1 of 8

What does the abbreviation DWB stand for?

- Deutsches Wörterbuch
- Deutsche Wortbuch
- Deutscher Wortschatz
- Deutsches Wortregister
- I don't know

Real-time opponent progress bar via Supabase Realtime



PvP — Elo Formula

Expected Score

$$E(A) = 1 / (1 + 10^{(R_B - R_A) / 400})$$

- **E(A)**: Expected probability that player A wins (0 to 1)
- **R_A, R_B**: Current Elo ratings of player A and player B

Rating update

$$R_{\text{new}} = R_{\text{old}} + K * (S - E)$$

- **R_new, R_old**: Player's rating after and before the match
- **K-factor**:
 - < 30 games played: K = 40 (volatile, lets new players find their level fast)
 - ≥ 30 games played: K = 32 (more stable)
- **S**: Actual result (1 = win, 0 = loss, 0.5 = draw)
- **E**: The expected result from the formula above (e.g., 0.5 means a 50/50 matchup)

THE RATING OF CHESSPLAYERS PAST & PRESENT

Second Edition

ARPAD E. ELO

Valuable Extensions

Users loved the platform but wanted more reasons to come back

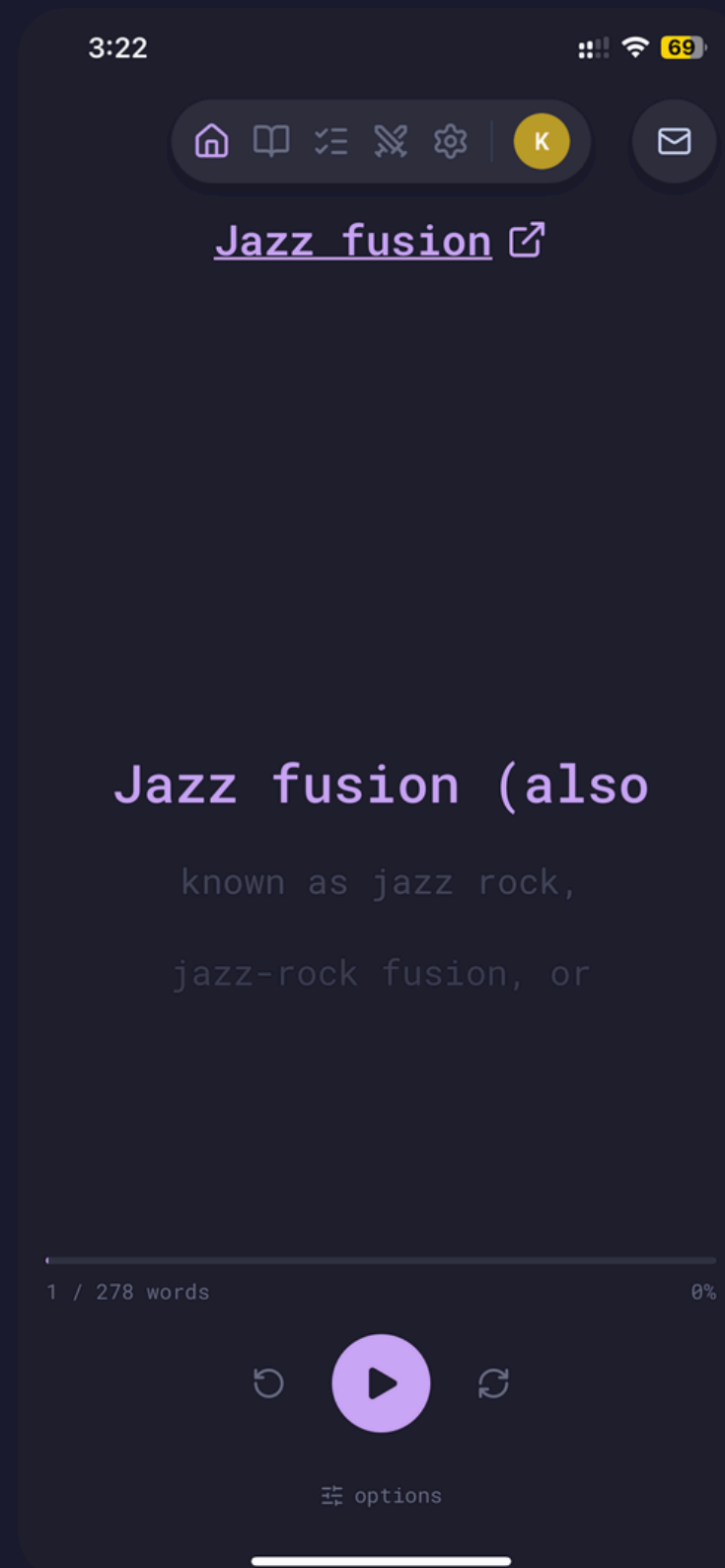
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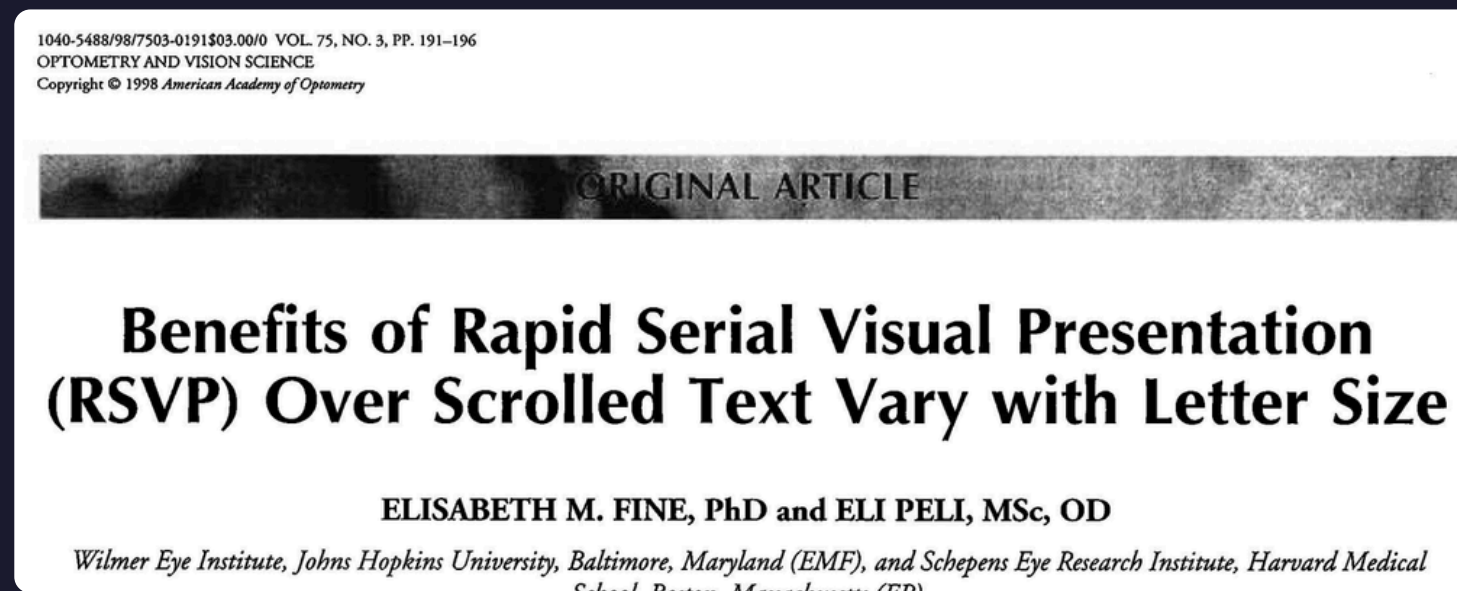
→ Enabled offline operations via PWA (Progressive Web Application)

Mobile Compatibility



RSVP Mode

Normally sighted observers read faster from an RSVP than a scroll display for all letter sizes tested.

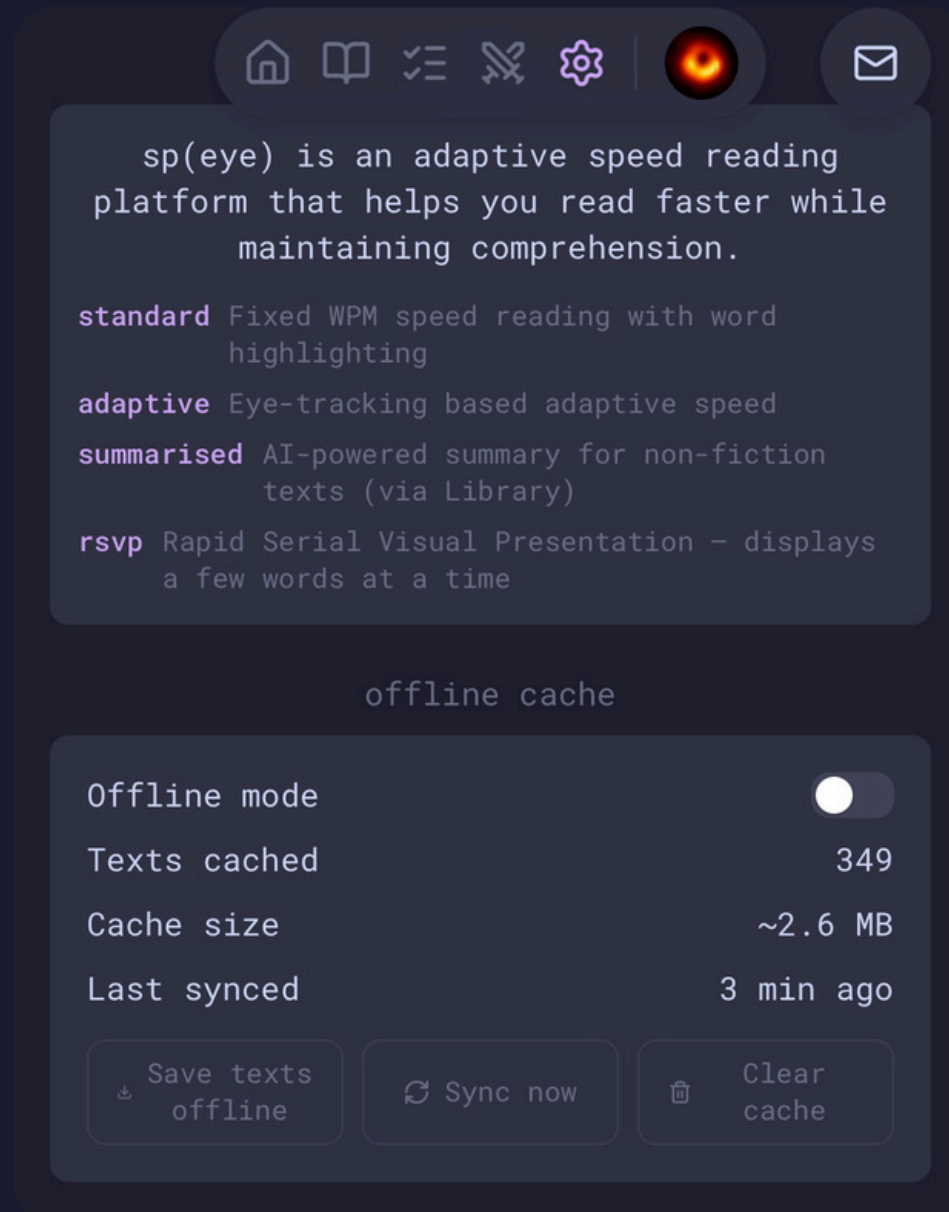


https://pelilab.partners.org/papers/Fine_OVS_1998.pdf

Mobile Compatibility

Progressive Web App

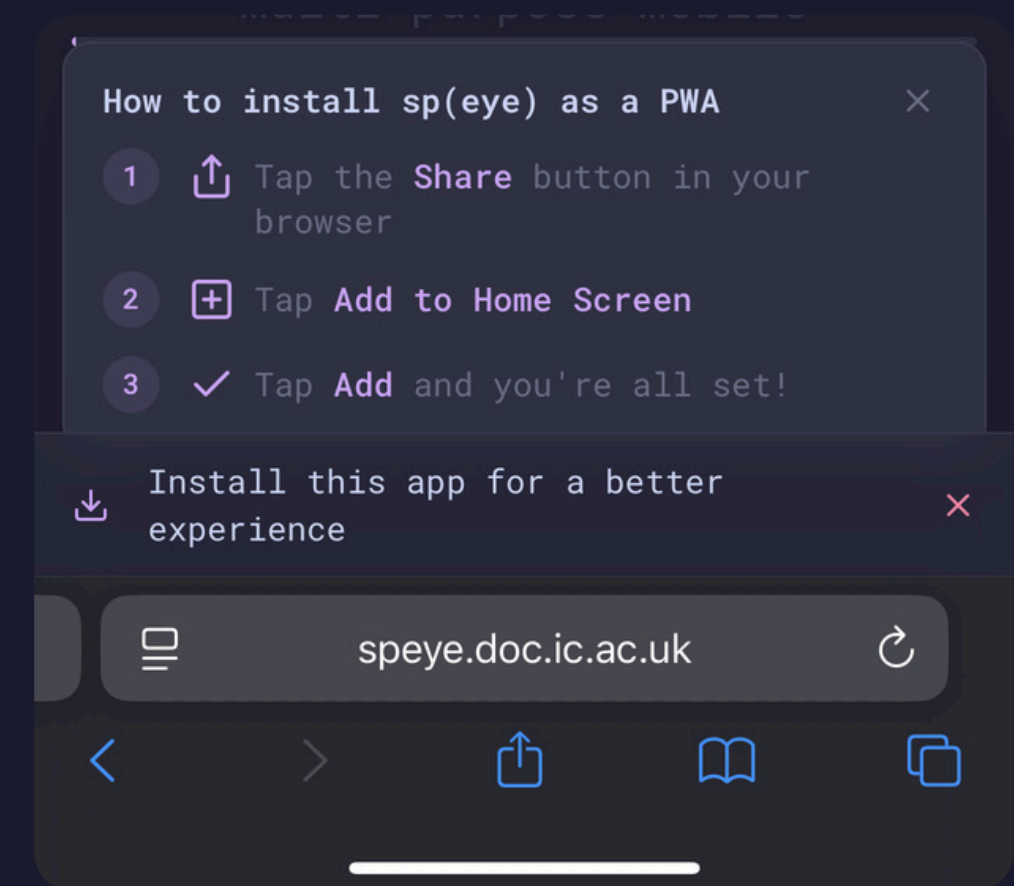
Easily installable; responsive and fast
Cost-effective cross-platform solution
Works offline via service workers

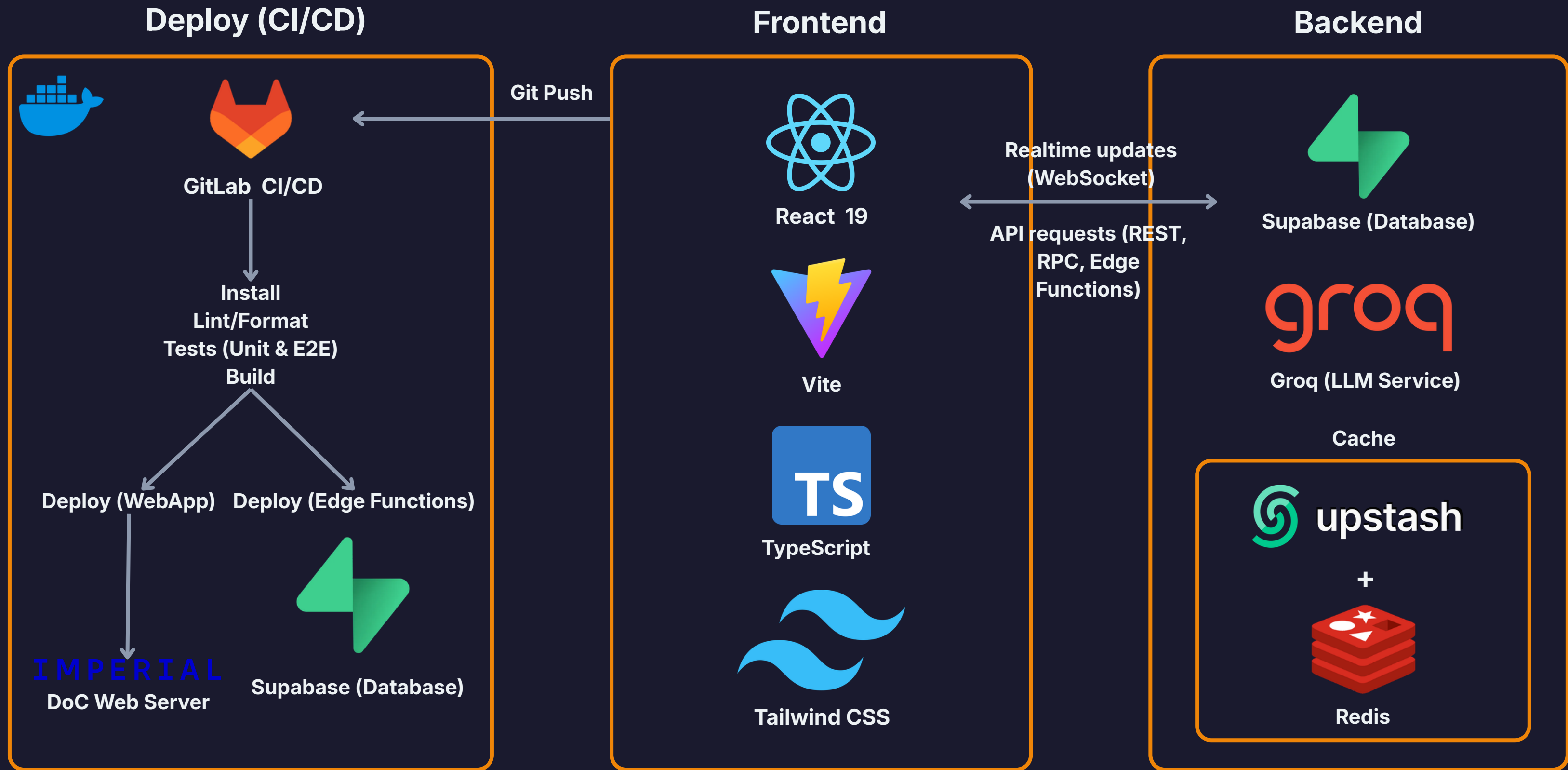


Offline Caching and syncing

Caches all texts into IndexedDB

Resyncs user activity automatically when next online





Evaluation: What went well?

Software Engineering Practices	• Agile (Scrum) approach with stand-ups, sprint planning, and retrospectives • Code reviews as standard practice before merging any PR
User testing	Feedback on UI/UX and reading experience

Evaluation: Future work

Strengths

Research driven
feature development

Extensive Testing
(Unit + Playwright E2E tests)

Scalability and
deployment readiness

User Testing

Limitations

Rate Limits

Insufficient time to
observe significant
improvements in
users' reading speed

Thank you for reading!

– Alexander Chow, Caleb Kan, Jeffrey Cheung, Kee Meng Tan, Royce Li Wei Chan, Ryan Ghoussainy